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WATER PURIFYING APPARATUS AND FILTER STRUCTURE

Abstract

A water purifying apparatus including a head having a water inlet portion and a water outlet portion; a shaft provided in the head and rotatably mounted between the water inlet portion and the water outlet portion; a bypass flow path passing through a circumference of the shaft, and communicating the water inlet portion with the water outlet portion according to rotation of the shaft; a filter detachably mounted to the head, and having a filter inserting portion in which the shaft is inserted when being mounted on the head; and a filtering flow path in which an inlet port and an outlet port are formed on the circumference of the shaft in a direction intersecting the bypass flow path, and communicated between the filter, the water inlet portion and the water outlet portion with each other according to the rotation of the shaft.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS [0001] This application is a continuation of U.S. application Ser. No. 17/874,012, filed on Jul. 26, 2022, which is a continuation of U.S. application Ser. No. 17/089,505, filed on Nov. 4, 2020, now U.S. Pat. No. 11,673,084, which is a continuation of U.S. application Ser. No. 16/116,759, filed on Aug. 29, 2018, now U.S. Pat. No. 10,905,987, which is a divisional of U.S. application Ser. No. 15/439,525, filed on Feb. 22, 2017, now U.S. Pat. No. 10,188,972, which claims the benefit of Korean Patent Application No. 10-2016-0034309, filed on Mar. 22, 2016. The disclosures of the prior applications are incorporated by reference in their entirety.

BACKGROUND

[0002] In general, a water purifying apparatus is used to purify water using a filter or a filtering material, and includes an apparatus that purifies water from a water pipe or tank.

[0003] A representative example of the water purifying apparatus that supplies drinkable water is a water purifier. Recently, the entire or a portion of such a water purifying apparatus has been provided even in a refrigerator to be supplied with purified water through the refrigerator or make ice using the purified water.

[0004] In a refrigerator including the water purifying apparatus, a filter constituting the water purifying apparatus is required to be replaced or inspected as necessary. Accordingly, for this purpose, the filter is detachably installed. The water purifying apparatus is configured to have a bypass flow path which is capable of supplying water even in a state where the filter is separated and then a user can dispense water even during replacement or inspection of the filter.

[0005] Korean Patent Laid-Open Publication No. 10-2015-0135021 discloses a head for a water purifying filter having a bypass structure. However, the head has a complicated structure to enable the flow path to be opened and closed by a flow path switching plunger which is supported by an elastic member during attachment and detachment of the filter. Therefore, there is decreased productivity and assembling workability. Additionally, the bypass path of the flow path is complicated due to the structure of the flow path switching plunger which is slid, thereby deteriorating the flow performance of water. Furthermore, the flow path is not capable of being switched when the performance of the elastic member is deteriorated or the elastic member is abnormal.

SUMMARY

[0006] The disclosed embodiments provide a water purifying apparatus which is capable of continuously supplying water by a bypass flow path being switched according to attachment and detachment of a filter.

[0007] The disclosed embodiments provide a water purifying apparatus which has a simple bypass flow path switching structure and thus is improved in productivity and assembling workability.

[0008] The disclosed embodiments provide a water purifying apparatus in which a bypass flow path is capable of being switched by rotation mounting of a filter.

[0009] The disclosed embodiments provide a water purifying apparatus which is easy the attachment and the detachment of the filter.

[0010] The disclosed embodiments provide a water purifying apparatus in which the flow of water is capable of being smoothly performed when the bypass flow path is converted by the attachment and the detachment of the filter.

[0011] According to one embodiment of the invention, a water purifying apparatus includes a head on which a water inlet portion and a water outlet portion are formed to face each other in a straight line; a mounting member on which a rotation supporting portion is formed to surround the water inlet portion and the water outlet portion of the head; a water inlet pipe and a water outlet pipe which are connected to the mounting member; a shaft provided in the head and rotatably mounted between the water inlet portion and the water outlet portion; a bypass flow path formed to pass through a circumference of the shaft, having a flow path projecting portion on a penetrated flow path, and communicating the water inlet portion with the water outlet portion according to rotation of the shaft; a filter detachably mounted to the head, and having a filter inserting portion in which the shaft is inserted when being mounted on the head; and a filtering flow path in which an inlet port and an outlet port are formed on the circumference of the shaft in a direction intersecting the bypass flow path, and which communicates between the filter, the water inlet portion and the water outlet portion with each other according to the rotation of the shaft.

[0012] According to another embodiment of the invention, a water purifying apparatus includes a head on which a water inlet portion and a water outlet portion are formed; a shaft provided in the head and rotatably mounted between the water inlet portion and the water outlet portion; a bypass flow path formed to pass through a circumference of the shaft and communicating the water inlet portion with the water outlet portion according to rotation of the shaft; a filtering flow path formed in the shaft in a direction intersecting the bypass flow path; an inner pipe formed to extend downward from the shaft and having a pipe cutout portion on an end thereof; a filter detachably mounted to the head and having a filter inserting portion in which the shaft is inserted when being mounted on the head; and a first connecting portion formed on a supporter extending portion which extends in a direction of the shaft in the filter insertion portion, to rotate the shaft upon rotatably coupling of the filter by coupling with the pipe cutout portion.

[0013] According to another embodiment of the invention, a water purifying apparatus includes a head on which a water inlet portion and a water outlet portion are formed; a shaft provided in the head, and rotatably mounted between the water inlet portion and the water outlet portion; a bypass flow path formed to pass through a circumference of the shaft and communicating the water inlet portion with the water outlet portion according to rotation of the shaft; a filtering flow path formed in the shaft to be separated from the bypass flow path; an inner pipe formed to extend downward from the shaft and having a pipe cutout portion on an end thereof and connected to the filtering flow path; a filter detachably mounted to the head and having a filter inserting portion in which the shaft is inserted when being mounted on the head; a supporter extending portion forming a first filter inlet flow path which is connected to the inner pipe and provided in the filter; and a first connecting portion formed on an upper portion of the supporter extending portion to rotate the shaft upon rotatably coupling of the filter by coupling with the pipe cutout portion.

[0014] According to another embodiment of the invention, a filter structure includes a housing

body formed in a cylindrical shape to have a first accommodating space and having an upper opening portion; a housing cap coupled to the upper opening portion of the housing body by a portion thereof being formed in a cylindrical shape to have a second accommodating space, wherein a coupling projection and an upper opening portion are formed on an outer circumferential surface thereof, and a housing fastening portion is formed on an inner circumferential surface thereof; a filtering member having a cylindrical shape accommodated in the first accommodating space; and an upper supporter coupled to an upper portion of the filtering member and an inner surface of the housing cap respectively, and accommodated in the second accommodating space.

[0015] The upper supporter includes a supporter accommodating portion coupled to the upper portion of the filtering member; a supporter stepped portion integrally formed on an upper portion of the supporter accommodating portion and in which a second filter inlet flow path is formed on the outer circumferential surface thereof so that water is supplied in a space between the outer circumferential surface of the filtering member and the housing body; a supporter extending portion which extends upward from the supporter stepped portion, in which a first filter inlet flow path which communicates with a second filter inlet flow path is formed on a center portion thereof, in which a pair of projecting portions are formed to be extend on the upper portion of the first filter inlet flow path and in which a filter outlet flow path is formed on an outside of the first filter inlet flow path; and a supporter fastening portion which prevents a rotation of the upper supporter in the housing cap by being coupled to the housing fastening portion.

[0016] Additional advantages, objects, and features of the invention will be set forth in part in the description which follows and in part will become apparent to those having ordinary skill in the art upon examination of the following or may be learned from practice of the invention. The objectives and other advantages of the invention may be realized and attained by the structure particularly pointed out in the written description and claims hereof as well as the appended drawings.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0017] The accompanying drawings, which are included to provide a further understanding of the invention and are incorporated in and constitute a part of this application, illustrate embodiments of the invention and together with the description serve to explain the principle of the invention. In the drawings:

[0018] FIG. 1 is a front view illustrating a refrigerator according to an embodiment of the invention.

[0019] FIG. 2 is a front view illustrating a state where a door of the refrigerator is opened according to an embodiment of the invention.

[0020] FIG. 3 is a view schematically illustrating a disposition structure of a water supplying flow path of the refrigerator according to an embodiment of the invention.

[0021] FIG. 4 is a perspective view illustrating the structure of the water purifying apparatus according to an embodiment of the invention.

[0022] FIG. 5 is a perspective view illustrating a structure of a water purifying apparatus according to another embodiment of the invention.

[0023] FIG. 6 is an exploded perspective view illustrating a state where a filter and a head of the water purifying apparatus are separated according to an embodiment of the invention.

[0024] FIG. 7 is an exploded view illustrating the structure of the filter according to an embodiment of the invention.

[0025] FIG. 8 is a longitudinal sectional view of FIG. 7.

[0026] FIG. 9 is a sectional view taken along line 9-9' of FIG. 7.

[0027] FIG. 10 is a sectional view taken along line 10-10' of FIG. 6.

[0028] FIG. **11** is a partially cutaway perspective view illustrating an upper supporter of the filter according to an embodiment of the invention.

[0029] FIG. **12** is a side view illustrating the head according to an embodiment of the invention.

[0030] FIG. **13** is an exploded perspective view illustrating a coupling structure of the head viewed from a side according to an embodiment of the invention.

[0031] FIG. **14** is an exploded perspective view illustrating the coupling structure of the head viewed from the other side according to an embodiment of the invention.

[0032] FIG. **15** is a cutaway exploded perspective view illustrating an internal structure of the head viewed from one side according to an embodiment of the invention.

[0033] FIG. **16** is a cutaway exploded perspective view illustrating the internal structure of the head viewed from the other side according to an embodiment of the invention.

[0034] FIG. **17** is a perspective view illustrating the head viewed from the lower side according to an embodiment of the invention.

[0035] FIG. **18** is a partially cutaway perspective view illustrating the head according to an embodiment of the invention.

[0036] FIG. **19** is a cutaway perspective view illustrating a state where the filter and the head are separated from each other according to an embodiment of the invention.

[0037] FIG. **20** is an enlarged view of portion A of FIG. **19**.

[0038] FIG. **21** is a view illustrating a shaft position in a state where the filter and the head are separated from each other according to an embodiment of the invention.

[0039] FIG. **22** is a view illustrating a state where the filter is inserted into an inside of the head according to an embodiment of the invention.

[0040] FIG. **23** is a view illustrating a state where the filter is fully inserted into the inside of the head according to an embodiment of the invention.

[0041] FIG. **24** is a view illustrating a state where the filter is rotated for coupling in a state where the filter is fully inserted into the inside of the head according to an embodiment of the invention.

[0042] FIG. **25** is an enlarged view of portion B in FIG. **24**.

[0043] FIG. **26** is a view illustrating a shaft position in a state where the filter is coupled to the head according to an embodiment of the invention.

DETAILED DESCRIPTION OF THE EMBODIMENTS

[0044] Hereinafter, some embodiments of the present disclosure will be described in detail with reference to the exemplary drawings. In the following description, the same elements will be designated by the same reference numerals although they are shown in different drawings. Further, in the following description of embodiments of the present disclosure, a detailed description of known functions and configurations incorporated herein will be omitted when it may make the subject matter of the present disclosure rather unclear.

[0045] FIG. **1** is a front view illustrating a refrigerator according to an embodiment of the invention. FIG. **2** is a front view illustrating a state where a door of the refrigerator of FIG. **1** is opened.

[0046] With reference to FIGS. **1** and **2**, an outer appearance of a refrigerator **1** which includes a water purifying apparatus may be formed by a cabinet **10** that forms a storage space, and a door **10** that opens and closes the storage space of the cabinet **10**.

[0047] The cabinet **10** may include an outer case **11** made of a metal material (not limited thereto) forming the outer surface and an inner case **12** made of a resin material (not limited thereto), which is coupled with the outer case **11** and forming a storage space in an inside portion of the refrigerator **1**. Insulation material may be provided between the outer case **11** and the inner case **12** to insulate the space in the inside of the refrigerator **1**.

[0048] The storage space may be separated in a vertical direction. The separation may be formed by a barrier **13**. Accordingly, the storage space may include an upper refrigerating compartment **14** and a lower freezing compartment **15**. The freezing compartment **15** may be further separated in a

lateral direction. It will be apparent that the storage space may be divided into left and right compartments based on the barrier **13**, and is not limited to any particular configuration.

[0049] The door **20** may include a refrigerating compartment door **21** and a freezing compartment door **22**, which independently open and close the refrigerating compartment **14** and the freezing compartment **15**, respectively.

[0050] Both of the refrigerating compartment door **21** and the freezing compartment door **22** may open and close the refrigerating compartment **14** and the freezing compartment **15** by rotation thereof. For this, each of the refrigerating compartment door **21** and the freezing compartment door **22** may be rotatably connected to the cabinet **10** by a door hinge **26** (see FIG. 3). In addition, the refrigerating compartment door **21** may be configured as a French type door configured such that a pair of doors independently rotate at both left and right sides.

[0051] A dispenser **23** and an ice maker **24** may be provided at one of the pair of refrigerating compartment doors **21**.

[0052] The dispenser **23** may be provided at a front surface of the refrigerating compartment door **21**, and enable a user to dispense water or ice by manipulating the dispenser **23** from the outside. An ice making compartment **25** may be provided above the dispenser **23**.

[0053] The ice making compartment **25** is a heat insulating space in which ice is made and stored. The ice maker **24** may be accommodated in inside the ice making compartment. The ice making compartment **25** may be opened/closed by a separate door. Although not shown in the drawings, the ice making compartment **25** may communicate with the freezing compartment **15** by a cool air duct in a state in which the refrigerating compartment door **21** is closed, and may receive cool air necessary for the ice making from a freezing compartment evaporator (not shown).

[0054] A plurality of shelves and drawers for storing food may be provided inside the refrigerating compartment **14**. For example, as shown in FIG. 2, a drawer assembly **16** may be provided at a bottom surface of the refrigerating compartment **14**. The drawer assembly **16** may include a drawer **161** that is slidable and a cover **162** that shields a top surface of the drawer **161**.

[0055] The drawer assembly **16** may be configured such that the inside thereof is visible therethrough, and a main water tank **34** (see FIG. 3) provided at a rear side of the refrigerating compartment **14** may be shielded by the drawer **161**.

[0056] A water purifying apparatus **17** may be provided at a side of the drawer assembly **16** for purifying water to be supplied and then supplying the purified water to the dispenser **23** and the ice maker **24**. The water purifying apparatus **17** may be disposed between the accommodating space and the wall surface of the drawer **161** and may be shielded by a front surface of the drawer **161**. Therefore, in a state where the drawer **161** is closed, the water purifying apparatus **17** is not exposed to the outside, and in a state where the drawer **161** is withdrawn, the water purifying apparatus is exposed to the outside thus providing access to the water purifying apparatus **17**. It is understood that the mounting position of the water purifying apparatus **17** is not limited to one side of the drawer **161** and may be provided in the region of the refrigerating compartment **14** including the refrigerating compartment **14** or the refrigerating compartment door **21**.

[0057] A plurality of shelves having a cantilever structure may be detachably provided above the drawer assembly **16** such that their heights are respectively adjustable. A main duct **18** may be provided at a rear surface of the refrigerating compartment **14** and cool air generated from an evaporator (not shown) may be supplied to the inside portion of the refrigerating compartment **14** through a plurality of discharging ports which are formed in a main duct **18**.

[0058] FIG. 3 is a view schematically illustrating a disposition structure of a water supplying flow path of the refrigerator according to an embodiment of the invention.

[0059] As shown in FIG. 3, the refrigerator **1** may include a water supplying flow path **30** that purifies or cools water being supplied from an external water supply source and then dispenses water from the dispenser **23**, or supplies the purified water to the dispenser **23** or the icemaker **24**.

[0060] The water supplying flow path **30** may be directly connected to a water supply source **2**,

such as a water supply pipe, at the outside of the refrigerator, and be introduced into a space of the refrigerator through a tube guide **19** mounted in the cabinet **10** to be connected to an inlet portion of the water purifying apparatus **17** in the refrigerator.

[0061] The water supplying flow path **30** may include a water supply valve **31** and a flow rate sensor **32**. The flow rate sensor **32** may be integrally formed with the water supply valve **31**.

[0062] The water supplying flow path **30** may connect the water purifying apparatus **17** and a first branch pipe **33** with each other. The water supplying flow paths **30** branched from the first branch pipe **33** may be connected to the main water tank **34** and a first branch valve **35**, respectively.

[0063] The water supplying flow path **30** which is connected to an outlet portion of the first branch valve **35** may extend along an upper surface after extending along a side wall of the inside of the cabinet **10** or a rear wall surface of the outside of the cabinet **10** through the tube guide **19** and may be introduced into the refrigerating compartment door **21** via the door hinge **26**.

[0064] The water supplying flow path of the refrigerating compartment door **21** may be branched by a second branch pipe **36** and connected to an inlet portion of a sub water tank **37** and a second branch valve **38**. The sub water tank **37** may be connected to the dispenser **23** so that cooled water is capable of dispensed through the dispenser **23**.

[0065] An outlet portion of the second branch valve **38** may be respectively connected to the dispenser **23** and the ice maker **24** by the water supplying flow path **30** to be capable of supplying purified water to the dispenser **23** and the ice maker **24**.

[0066] The water purified through the water purifying apparatus **17** may be cooled and then supplied to the dispenser **23** or may be supplied to the dispenser **23** or the ice maker **24** in a state in which the water is purified but not cooled.

[0067] The water purifying apparatus **17** may include a plurality of filters **40** for purifying supplied water and a head unit **50** to which the plurality of filters **40** are coupled and which is connected a flow path through which water flows. The water purifying apparatus **17** may include a case **171** that accommodates the filters **40** and the head unit **50**.

[0068] FIG. **4** is a perspective view illustrating the structure of the water purifying apparatus according to an embodiment of the invention.

[0069] As shown in FIG. **4**, the water purifying apparatus **17** may include a plurality of filters **40** and a head unit **50**.

[0070] The plurality of filters **40** may include a first filter **401** which is connected to a water inlet side of the head unit **50**, a third filter **403** which is connected to a water outlet side of the head unit **50**, and a second filter **402** disposed between the first filter **401** and the third filter **403** and thus is capable of purifying water. The first filter **401** may be a pre-carbon filter, the second filter **402** may be a membrane filter, and the third filter **403** may be a post-carbon filter.

[0071] It is understood that the invention is not limited to the number and types of the filters **40**, and the number which is capable of being accommodated in the inside of the water purifying apparatus **17** and different types of functional filters from each other for effectively purifying water may be applied.

[0072] The head unit **50** may include a plurality of heads **60** to which each of the filters **40** is coupled and a mounting member **70** on which the head **60** is rotatably seated.

[0073] A water inlet pipe **301** for supplying original water is capable of being connected to one end of the mounting member **70**, and a water outlet pipe **302** for discharging purified water is capable of being connected to the other end thereof.

[0074] Each of the plurality of heads **60** may be independently rotated in a state where the plurality of heads **60** are mounted on the mounting member **70**. The flow paths of the plurality of heads **60** are capable of being connected to each other by connecting pipes **71** and the original water which flows in through the water inlet pipe **301** is purified after passing through each of the filters **40** and then may be flows out from the water outlet pipe **302**.

[0075] Each of the connecting pipes **71** may be mounted on the mounting member **70** and is

provided between adjacent two heads **60** to each other to allow water to flow between each of the heads **60**. A cover **72** may be mounted on one side of the mounting member **70** which corresponds to the connecting pipe **71** to shield the connecting pipe **71**.

[0076] The head **60** may include a head body **61** to which an upper end of the filter **40** is inserted and then fixed, and a shaft (**90** in FIG. **6**) connected to the upper end of the filter **40** in the inside portion of the head body **61** and thus forms a flow path through which water flows. The head **50** may further include a head cap **62** for shielding an upper surface of the head body **61** into which the shaft **90** may be inserted.

[0077] The filter **40** may be fixedly mounted to the head **60** in a rotatable manner. The shaft **90** may be connected to the filter **40** to form a flow path in a process of the filter **40** being mounted and when the filter **40** is rotated, the shaft **90** is rotated along with the filter **40** so that the flow path of the shaft **90** is capable of being switched.

[0078] For example, in a case where the filter **40** is mounted, the flow path may be switched to the filter **40** side by the shaft **90** so that the water is capable of being purified through the filter **40**. In a case where the filter **40** is separated, the supplying water is capable of being bypassed without passing through the filter **40**, and thus the flow path is capable of being switched so that water passes through the head **60**. The switching and detailed structure of the flow path will be described in more detail in the following other embodiments.

[0079] The case **171** may have various structures which can accommodate the filter **40** and the head unit **50**. The case **171** is capable of fully accommodating the filter **40** and the head unit **50**. Alternatively, the case **171** may accommodate a portion of the filter **40** and/or the head unit **50**.

[0080] The case **171** may have a structure which is capable of being fixedly mounted at one side of the inside portion of the refrigerating compartment **14**. Alternatively, the case **171** may not be provided and the mounting member **70** may be directly mounted to a side of the inner portion of the refrigerating compartment **14**.

[0081] Only one filter **40** may be provided according to the function of the water purifying apparatus **17**. In such case where only one filter **40** is provided, one head unit **50** and one mounting member **70** may be provided. The structure of the head **60** may have the same structure irrespective of whether there is just one head **60** or there are a plurality of the heads **60**. Hereinafter, a water purifying apparatus having one filter **40** and one head **60** will be described.

[0082] FIG. **5** is a perspective view illustrating a structure of a water purifying apparatus according to another embodiment of the invention. FIG. **6** is an exploded perspective view illustrating a state where a filter and a head of the water purifying apparatus are separated.

[0083] With reference to FIGS. **5** and **6**, a water purifying apparatus **17** may include a filter **40** and a head **60**. The water purifying apparatus **17** may further include a mounting member **70** on which the head **60** is mounted.

[0084] The filter **40** may have a cylindrical shape (not limited thereto). The outer shape of the filter **40** may be formed by the housing **41**. The housing **41** may include a housing body **42** to accommodate a filtering member (**44** in FIG. **7**) in an inside portion thereof, and a housing cap **43** which may be coupled to an upper end of the housing body **42** to form an upper portion of the housing **41**.

[0085] The housing body **42** may have a cylindrical shape (not limited thereto) to define a first accommodating space for accommodating the filtering member **44**. The housing body **42** may have an upper opening portion.

[0086] The housing cap **43** may be coupled to the upper opening portion of the housing body **42**. The housing cap **43** may define a second accommodating space for accommodating a portion of the filtering member **44**. As such, a portion of the housing cap **42** may have a cylindrical shape. The housing cap **43** has a top hollow space **435** having an open lower end thereof communicating with the second accommodating space. A portion of the shaft **90** is capable of being inserted through the upper opening portion of the housing cap **43** (discussed in more detail below).

[0087] The housing cap **43** may be inserted into an opened lower surface of the head **60**. A pair of O-rings **432** may be provided at an upper end of the housing cap **43**. The O-rings **432** may be hermetically sealed with the inner surface of the head **60** to prevent leakage.

[0088] A coupling projection **433** may be provided on an outer circumferential surface of the upper portion of the housing cap **43**. The coupling projection **433** is capable of being moved along a coupling groove **631** which is formed on an inner surface of the head **60** when the upper portion of the filter **40** is inserted into the inside of the head **60**.

[0089] The coupling projection **433** and the coupling groove **631** may be formed in a direction intersecting a direction into which the filter **40** is inserted. Accordingly, the filter **40** is rotated in a state of being inserted into the inside of the head **60** and may have a structure in which the coupling projection **433** and the coupling groove **631** are coupled to each other by rotation of the filter **40**. In a state where the filter **40** and the head **60** are fully coupled to each other, the supplying water is capable of flowing into the inside portion of the filter **40** by the filter **40** and the flow path of the head **60** being connected to each other in the inside portion thereof.

[0090] The mounting member **70** may include a base **73** which is mounted to one side wall surface of the refrigerator **1** or the case **171**, and rotating support portions **74** which project from both sides of the base **73** and rotatably support both sides of the head **60**.

[0091] End portions of the water inlet pipe **301** and the water outlet pipe **302** may be disposed on the rotating support portion **74**. The water inlet pipe **301** and the water outlet pipe **302** may be connected to a water inlet portion **611** and a water outlet portion **612** of the head **60** at the rotating support portion **74**.

[0092] The head **60** may be rotatably mounted on the mounting member **70** by the rotating support portion **74**. Therefore, a space for attachment and detachment of the filter **40** can be formed by operating rotation or tilting of the head **60** at the time of attachment and detachment of the filter **40**. Thus, the manual operation for the attachment and detachment of the filter **40** can be more easily performed.

[0093] The head **60** may be formed in a cylindrical shape (not limited thereto) having an opened lower surface. The head **60** may include a head body **61** to which the filter **40** is inserted and fixed, and a shaft **90** which is accommodated in the inside of the head body **61**. The head **60** may further include a head cap **62** to shield an opened upper surface of the head body **61**.

[0094] An insertion indicating portion **613** for indicating an insertion position of the coupling projection **433** may be formed on the outer surface of the head body **61**. The insertion indicating portion **613** may be formed by printing, molding or machining. The user can recognize the position of the coupling groove **631** by viewing the insertion indicating portion **613**, and may more easily perform alignment between the coupling projection **433** and the coupling groove **631**.

[0095] A rotation indicating portion **614** for indicating the rotational direction of the filter **40** may be formed on the outer surface of the head body **61**. The rotation indicating portion **614** may be formed by printing, molding or machining. The engagement projection **433** may be moved along and coupled to the inside of the coupling groove **631** by the user operating rotation of the filter **40** in a correct direction by the rotation indicating portion **614**.

[0096] An opening portion **632** for checking the engaging restraint state of the coupling projection **433** may be further formed on the outer surface of the head body **61**. The opening portion **632** may be formed at a position which corresponds to the position of the coupling groove **631** or may include at least a portion of the coupling groove **631**.

[0097] FIG. **7** is an exploded view illustrating the structure of the filter according to an embodiment of the invention. FIG. **8** is a longitudinal sectional view of FIG. **7**. FIG. **9** is a cross-sectional view taken along line **9-9'** of FIG. **7**. FIG. **10** is a cross-sectional view taken along line **10-10'** of FIG. **6**.

[0098] With reference to FIGS. **7-10**, the filter **40** may include a filter housing **41** which forms an outer shape of the filter **40**, a filtering member **44** provided in the filter housing **41**, and an upper

supporter **80** for supporting the filtering member **44**.

[0099] The filter **40** may further include a lower supporter **45** for supporting the filtering member **44** in the filter housing **41**.

[0100] The filter housing **41** may be formed in a cylindrical shape (not limited thereto), and may be formed by the housing body **42** and the housing cap **43** being coupled together. A filter inserting portion **431** may be formed on the upper end of the housing cap **43**. A plurality of O-rings **432** may be vertically arranged in the filter inserting portion **431**.

[0101] The coupling projection **433** may be formed on a lower side of the plurality of O-rings **432** on the outside of the filter inserting portion **431**. A pair of coupling projections **433** may be formed at positions which are opposite to each other, and may be formed to have a size that is capable of being inserted into the coupling groove **631**.

[0102] The coupling projection **433** may include a projection guide portion **433a**. The projection guide portion **433a** may be formed to have a slope or a predetermined curvature on the upper surface of the coupling projection **433**. The coupling projection **433** is configured to be in contact with a groove guide portion (**633** in FIG. 15) for guiding the coupling projection **433** to the entrance of the coupling groove **631** so as to guide rotation of the coupling projection **433** in one direction.

[0103] A restraining projection **433b** which projects to a lower side may be formed on one side of the lower surface of the coupling projection **433**. The restraining projection **433b** may be engaged with the inside of the coupling groove **631** to be restrained. Accordingly, the filter **40** is capable of being fixed to the inside of the head **60** in a state where the coupling projection **433** is fully inserted into the inside of the coupling groove **631**.

[0104] On the other hand, the inside portion of the filter inserting portion **431** may be formed having a hollow shape. A housing fastening portion **434** for being coupled with the upper supporter **80** may protrude from the inner side of the filter inserting portion **431**. In other words, the upper supporter **80** is capable of being accommodated in the second accommodating space and the top hollow space **435** which are defined in the housing cap **43**. The upper supporter **80** may be coupled to the upper portion of the filtering member **44** and may be received in the top hollow space **435** in the housing cap **43**, respectively.

[0105] The upper surface **434a** of the housing fastening portion **434** may be an inclined surface. For example, a pair of housing fastening portions **434** may be formed at positions facing each other, and may be formed so that inclined directions thereof are opposed to each other.

[0106] A supporter fastening portion **81** may be formed on the upper supporter **80** which corresponds to the housing coupling portion **434**. For example, the supporter fastening portion **81** may be formed on the second extending portion **87**. A hook portion **811** may be formed at one end of the supporter fastening portion **81**.

[0107] In a case where the upper end of the upper supporter **80** may be rotated in one direction in a state of being inserted into the filter inserting portion **431**, the hook portion **811** of the supporter fastening portion **81** moves along the inclined surface of the upper surface of the housing fastening portion **434** to be restrained. In a state where the upper supporter **80** is fully rotated, the hook portion **811** of the supporter fastening portion **81** engages with the end portion of the housing fastening portion **434** and thus the upper supporter **80** is capable of being coupled to the housing cap **43**.

[0108] The supporter fastening portion **81** and the housing fastening portion **434** may have various structures which are mutually combined with each other and the filter housing **41** and the upper supporter **80** in the inside portion of the filter housing **41** may be integrated with each other by the supporter fastening portion **81** and the housing fastening portion **434** being coupled to each other. Therefore, when the filter **40** is rotated while the filter **40** is being mounted, the upper supporter **80** may also rotate with the filter **40**.

[0109] On the other hand, the filtering member **44** may be accommodated inside the filter housing

41. The filtering member **44** may be a standard carbon filter or a membrane filter. However, it is understood that various types of filters may be used depending on the required purifying performance.

[0110] The filtering member **44** may be formed in a cylindrical shape (not limited thereto) having a hollow **441** formed at the center thereof in the vertical direction. The upper supporter **80** and the lower supporter **45** may be coupled to the upper end and the lower end of the filtering member **44** so that the filtering member **44** is capable of being fixedly mounted on the inside of the filter housing **41**.

[0111] An outer diameter of the filtering member **44** may be smaller than the inner diameter of the filter housing **41** and a space for flowing water between the filter housing **41** and the outer surface of the filtering member **44** may be formed.

[0112] The upper supporter **80** may be disposed at the upper end of the filtering member **44** and extend in the upper direction to form a passage connecting the inlet portion of the filter inserting portion **431** and the hollow **441** to each other. Accordingly, water supplied from the head **60** is capable of flowing into the filtering member **44** through the filter inserting portion **431** and water purified in the filtering member **44** flows out to the head **60**.

[0113] The upper supporter **80** may include a supporter accommodating portion **82** for accommodating the upper end of the filtering member **44**.

[0114] The upper supporter **80** may further include a supporter inserting portion **83** which extends from a central portion of the supporter accommodating portion **82** in the lower direction and is inserted into the hollow **441** of the filtering member **44**.

[0115] The upper supporter **80** may further include a supporter stepped portion **84** which projects from the upper surface of the supporter inserting portion **83** in the upper direction.

[0116] The upper supporter **80** may further include a supporter extending portion **85** extending from the center of the upper surface of the supporter stepped portion **84** toward the inside of the filter inserting portion **431**.

[0117] FIG. **11** is a partially cutaway perspective view illustrating an upper supporter of the filter according to an embodiment of the invention.

[0118] The upper supporter **80** will be described in more detail with reference to FIGS. **8-11**.

[0119] When the upper supporter **80** and the filtering member **44** are coupled together, the supporter accommodating portion **82** may surround the upper surface and a circumference of the filtering member **44**. The supporter inserting portion **83** may be inserted into the hollow **441** such that it is in contact with an inner surface of the filtering member **44** so that the upper supporter **80** is capable of being fixedly mounted on the upper surface of the filtering member **44**.

[0120] The inside portion of the supporter inserting portion **83** may be hollow such that it is capable of communicating with a filter outlet flow path **861** formed on the supporter extending portion **85**. Accordingly, the purified water flowing into the hollow **441** of the filtering member **44** may pass through the supporter inserting portion **83**, the filter outlet flow path **861** and a filter outlet port **862** of the end portion of the filter outlet flow path **861** in this order and be discharged through the opening of the filter inserting portion **431**.

[0121] The supporter stepped portion **84** may protrude from the upper surface of the supporter accommodating portion **82**. The supporter stepped portion **84** may have a smaller diameter than the supporter accommodating portion **82**. The circumference of the supporter stepped portion **84** and the supporter accommodating portion **84** may be spaced apart from the inner surface of the housing cap **43** when the housing cap **43** and the upper supporter **80** are coupled together.

[0122] The supporter extending portion **85** may extend from the center of the supporter step portion **84** in the upper direction. The supporter extending portion **85** may be located in the filter inserting portion **431** when the housing cap **43** and the upper supporter **80** are coupled together. A filter inlet flow path **871** and a filter outlet flow path **861** may be formed in the inside of the supporter extending portion **85**. Therefore, water supplied into the inside portion of the filter **40** and

discharged from the purified water from the filter **40** is capable of being performed through the supporter extending portion **85**.

[0123] For example, a first filter inlet flow path **872** extending in a lower direction may be formed on the opened upper surface of the supporter extending portion **85**. The supporter stepped portion **84** may include a second filter inlet flow path **873** passing across the supporter stepped portion **84**.

[0124] The second filter inlet flow path **873** may have an opening formed in a circumferential surface of the supporter stepped portion **84** and extending toward the center of the supporter stepped portion **84**. The first filter inlet flow path **872** and the second filter inlet flow path **873** may be connected to each other at a lower end of the supporter extending portion **85**, that is, at the inside of the supporter stepped portion **84**.

[0125] Accordingly, the water which flows through the filter inserting portion **431** flows through the first filter inlet flow path **872** of the supporter extending portion **85** and then flows along the second filter inlet flow path **873** which is branched into both sides at the lower end of the first filter inlet flow path **872** to the outside and is thus capable of being discharged through an opening of the circumference of the supporter stepped portion **84**.

[0126] The water which is discharged through the filter inlet flow path **871** flows along the space between the filter housing **41** and the filtering member **44**. The water which flows into the outside of the filtering member **44** may be purified in the process of coming in the hollow **441** through the filtering member **44**.

[0127] On the other hand, the supporter extending portion **85** may include a first extending portion **86** which extends from the upper surface of the supporter stepped portion **84** and a second extending portion **87** which extends from the first extending portion **86** in the upper direction. The first extending portion **86** may have an outer diameter which is larger than the outer diameter of the second extending portion **87** and the filter outlet port **862** may be formed on the upper side portion of the first extending portion **86**.

[0128] A guide surface **874** which is recessed in the inside direction may be formed on the outer surface of the second extending portion **87** of the upper side of the filter outlet port **862**. The guide surface **874** may be inclined and spaced apart from the filter outlet port **862** in the upper direction. Therefore, when the purified water which is discharged to the filter outlet port **862** flows in the upper direction, the water is guided to the outside of the second extending portion **87** by the guide surface **874** so that the water can flow more effectively.

[0129] A portion of the second extending portion **87** may extend toward the inside of the first extending portion **86** at the lower end of the guide surface **874**. At this time, the filter outlet flow path **861** may be defined in a space between an inner surface of the first extending portion **86** and an outer surface of the second extending portion **87**. The first filter inlet flow path **872** may extend from the opened upper surface of the second extending portion **87** to the lower end thereof.

[0130] On the other hand, the outer diameter of the first extending portion **86** is formed to be somewhat smaller than the inner diameter of the filter inserting portion **431** and the purified water which is discharged through the filter outlet port **862** is capable of being discharged through a space between the first extending portion **86** and the filter inserting portion **431**.

[0131] An outlet port groove **875** may be formed in the lower end of the guide surface **874** of the upper side of the filter outlet port **862**. The outlet port groove **875** may be formed to be recessed along the circumference of the second extending portion **87**. Accordingly, the water which is discharged from the filter outlet ports **862** is capable of flowing along the outlet port groove **875** and being discharged along the water outlet guide portion **965b** formed on the outer surface of the shaft **90**.

[0132] The outer diameter of the second extending portion **87** may correspond to the inner diameter of the shaft **90** which is located in the inside of the filter inserting portion **431**. Therefore, in a case where the second extending portion **87** and the shaft **90** are communicably coupled together, water supplied through the inside of the shaft **90** is capable of flowing inside of the second extending

portion **87**.

[0133] A first extending portion O-ring **863** may be provided on the outside of the first extended portion **86** and a second extending portion O-ring **876** may be provided on the outside of the second extending portion **87**. In this connection, a first receiving groove **864** may be inwardly defined in the outer face thereof, wherein the groove **864** accommodates therein the first sealing member **863**. The second extending portion O-ring **876** may be provided on the outer face of the support **80**, wherein the first extending portion O-ring **863** is disposed below the second extending portion O-ring **876** and is spaced therefrom. The second extending portion O-ring **876** contacts the supporter **80** and is spaced from the inner face of the housing cap **43**. A second receiving groove **877** may be inwardly defined in the outer face thereof. The groove **877** accommodates therein the second sealing member **876**. The first extending portion O-ring **863** is provided to create a hermetic seal between the inside of the filter inserting portion **431** and the first extending portion **86**. The second extending portion O-ring **876** may be in close proximity with the inner surface of the shaft **90** to provide a hermetic seal between the second extending portion **87** and the shaft **90**.

[0134] Therefore, the purifying water which flows into the inside of the supporter extending portion **85** and the purified water which is discharged to the outside of the supporter extending portion **85** may flow through independent flow paths respectively without leakage or mixing with each other (or substantially reduced leakage/mixing).

[0135] On the other hand, a first connecting portion **851** may be formed on the upper end of the supporter extending portion **85**. The first connecting portion **851** may be recessed toward the inside from the upper end of the supporter extending portion **85** and may be symmetrical to both the left side and the right side thereof. The first connecting portion **851** may have a shape corresponding to that of the second connecting portion **972** of the shaft **90** (to be described below) so as to be engaged with each other when the second connecting portion **972** is inserted into the first connecting portion **851**. The shaft **90** and the filter **40** may be rotated together in a state where the first connecting portion **851** and the second connecting portions **972** are coupled together.

[0136] The first connecting portion **851** may be symmetrical with respect to a projecting portion **852** projecting from a position which is opposite to the inner surface of the supporter extending portion **85** while being recessed at a predetermined depth. A first inclined surface **853** may be formed on both side ends of the first connecting portion **851**, that is, on both side surfaces of the projecting portion **852**. The first inclined surface **853** may be shaped such that the width of the first connecting portion **851** gradually increases and the width of the projecting portion **852** gradually decreases from the lower side to the upper side of the first inclined surface. In other words, the first connecting portion **851** may be formed so that the width thereof is gradually narrowed toward the depth direction to be recessed.

[0137] The first inclined surface **853** may be inclined in a direction crossing a lengthwise vertical direction of the first inclined surface **853**, that is, in a direction of rotation of the filter **40**. Accordingly, the projecting portion **852** may be formed so that the width thereof is gradually widen from the inner surface of the supporter extending portion **85** toward the center of the inner diameter of the supporter extending portion **85**.

[0138] With such configuration, the first connecting portion **851** and the second connecting portion **972** can be more easily assembled. In addition, in a case where the torsion moment is applied when the first connecting portion **851** and the second connecting portion **972** are in contact with each other, the first connecting portion **851** and the second connecting portion **972** are slid so that the filter **40** can be more easily separated from the shaft **90**.

[0139] FIG. **12** is a side view illustrating the head according to an embodiment of the invention. FIG. **13** is an exploded perspective view illustrating a coupling structure of the head viewed from a side, and FIG. **14** is an exploded perspective view illustrating the coupling structure of the head viewed from the other side.

[0140] With reference to FIGS. **12-14**, the shaft **90** may be inserted through the opened upper

surface of the head body **61**, and the head cap **62** may shield or cover the opened upper surface of the head body **61**.

[0141] The head body **61** may include a lower body **63** and an upper body **64**.

[0142] The lower body **63** is a portion into which the filter inserting portion **431** may be inserted and to which the filter inserting portion **43** may be coupled. The lower surface of the lower body **63** may be opened so that the filter **40** can be accommodated. The coupling groove **631** for inserting the coupling projection **433** may be formed on the lower body **63**.

[0143] An opening portion **632** for forming the coupling groove **631** may be formed on the lower body **63** and the insertion state of the coupling projection **433** may be confirmed through the opening portion **632**. A plurality of supporting ribs **634** for supporting the outer side of the filter inserting portion **431** may be formed on the circumference of the inner surface of the lower body **63** to prevent the filter **40** from sagging in a state where the filter inserting portion **431** is inserted into the lower body **63**.

[0144] The upper body **64** may be formed at an upper end of the lower body **63** and may have a smaller diameter than the diameter of the lower body **63**. The shaft **90** may be mounted on the inside of the upper body **64** and the shaft **90** is inserted through the opened upper surface of the upper body **64** and thus may be mounted on the inside of the upper body **64**.

[0145] The upper end of the filter inserting portion **431**, the upper end of the supporter extending portion **85**, and the lower end of the shaft **90** may be disposed on the inside of the upper body **64**. A flow path through which purified water through the filter **40** is capable of flowing may be formed by coupling portions.

[0146] A water inlet portion **611** and a water outlet portion **612** may be formed to project to the outside in the upper body **64**. The water inlet portion **611** and the water outlet portion **612** may communicate with a water inlet pipe **301** and a water outlet pipe **302**, respectively. At this time, the water inlet portion **611** and the water outlet portion **612** can selectively communicate with the flow path formed in the shaft **90**. The water inlet portion **611** and the water outlet portion **612** may be disposed in a straight line so as to face each other in the head.

[0147] The head cap **62** may shield the upper surface of the upper body **64**. More particularly, the head cap **62** may press against the upper surface of the shaft **90** so that the shaft **90** can maintain a state of being fixedly mounted on the inside of the upper body **64**. To this end, a cap support portion **621** extending to the upper surface of the shaft **90** may be formed on a lower surface of the head cap **62**.

[0148] The shaft **90** is capable of selectively switching the flow path of water flowing in the inside of the head **60**. The shaft **90** may be rotatably seated on the upper body **64**. A filtering flow path **96** and a bypass flow path **95** may be formed in the shaft **90**. The flow paths may be selectively connected to the water inlet portion **611** and the water outlet portion **612** by the rotation of the shaft **90**.

[0149] The shaft **90** may include an upper part **91** and a lower part **92**. The upper part **91** may have a larger diameter than the diameter of the lower part **92**, and may have an outer diameter corresponding to the inner diameter of the upper body **64**.

[0150] A shaft O-ring **93** may be provided at the upper end and the lower end of the upper part **91**, respectively. The shaft O-ring **93** may provide a hermetic seal between the shaft **90** and the inner surface of the upper body **64** so as to prevent leakage of water flowing through the head **60**.

[0151] A shaft inlet port **961** and a shaft outlet port **962** may be formed on the circumference of the upper part **91** between the plurality of shaft O-rings **93**, respectively. More particularly, the shaft inlet port **961** and the shaft outlet port **962** may be formed at positions facing each other and at positions corresponding to the water inlet portion **611** and the water outlet portion **612**. Therefore, water which passes through the water inlet portion **611** can flow in the shaft inlet port **961**, and water which passes through the shaft outlet port **962** can be discharged through the water outlet portion **612**.

[0152] On the other hand, the shaft inlet port **961** and the shaft outlet port **962** may have a rectangular shape; however it is not limited to this. A sealing member mounting portion **911** on which a sealing member **94** is mounted may be formed on the circumference of the shaft inlet port **961** and the shaft outlet port **962**. The sealing member mounting portion **911** and the sealing member **94** may have a rectangular shape corresponding to the shaft inlet port **961** and the shaft outlet port **962**.

[0153] At this time, the sealing member **94** may include a pressing portion **941** which is pressed into and mounted on the inside of the sealing member mounting portion **911**, and a sealing portion **942** which projects along the circumference of the pressing portion **941**. The scaling portion **942** may be in contact with the inner surface of the upper body **64** when the shaft **90** is mounted. Accordingly, when the shaft inlet port **961** and the shaft outlet port **962** are connected to the water inlet portion **611** and the water outlet portion **612** by the rotation of the shaft **90**, leakage of water between the shaft **90** and the head **60** is prevented or substantially reduced. The shape of the sealing member **94** may be formed to be lengthened in a direction in which the shaft **90** is rotated so that the removal of the sealing member **94** from the shaft **90** or interference of the sealing member **94** with the shaft **90** is minimized during the rotation of the shaft **90**.

[0154] A scaling rib **613** may be formed in a water inlet port **611a** and a water outlet port **612a** which are formed in the water inlet portion **611** and the water outlet portion **612**, respectively. The sealing ribs **613** prevent the sealing member **94** from being separated or damaged and may be formed to cross the water inlet port **611a** and the water outlet port **612a** of the inner surface of the upper body **64**. The sealing ribs **613** may extend laterally and a plurality of sealing ribs **613** may be disposed in a direction intersecting each other.

[0155] Therefore, the sealing member **94** which is passed by the water inlet port **611a** and the water outlet port **612a**, which are formed in the water inlet portion **611** and the water outlet portion **612** respectively, is hooked into the water inlet port **611a** and the water outlet port **612a** in a process of the rotation of the shaft **90**. Thus, the sealing member **94** is capable of being prevented from being separated from the sealing member mounting portion **911**. In other words, since a state where the sealing ribs **613** press against the outer surface of the sealing member **94** in the process of the rotation of the shaft **90**, the sealing member **94** is capable of being prevented from being separated from the sealing member mounting portion **911**.

[0156] A bypass inlet port **951** and a bypass outlet port **952** may be formed between the shaft inlet port **961** and the shaft outlet port **962** of the outer surface of the upper part **91**, respectively. The bypass inlet port **951** and the bypass outlet port **952** may also be disposed at positions facing each other. Accordingly, in a case where the bypass inlet port **951** and the bypass outlet port **952** are aligned with the water inlet portion **611** and the water outlet portion **612** by the rotation of the shaft **90**, the water supplied to the water inlet portion **611** is capable of being discharged directly to the water outlet portion **612** through the bypass flow path **95**.

[0157] The bypass inlet port **951** and the bypass outlet port **952** may be formed at positions which are the same height as the shaft inlet port **961** and the shaft outlet port **962**, respectively, and which are rotated by 90 degrees to each other. Therefore, to connect the filtering flow path **96** or the bypass flow path **95** to the water inlet portion **611** and the water outlet portion **612**, the shaft **90** should be rotated by 90 degrees.

[0158] A portion which forms the upper end of the filtering flow path **96** is capable of projecting from the bypass flow path **95** in the inside of the upper part **91**. Accordingly, it is possible to realize the filtration flow path **96** and the bypass flow path **95** at the same time in a state where the height of the shaft **90** is minimized. Through this, the head **60** and the water purifying apparatus **17** can have a more compact configuration.

[0159] A rotating projection **921** may be provided on the lower surface of the upper part **91**. A pair of rotating projections may be formed and one rotating projection may be formed at a position corresponding to the shaft inlet port **961** and the shaft outlet port **962**, respectively. In other words,

the pair of rotating projections **921** may be formed at positions which are rotated by 90 degrees with respect to each other.

[0160] On the other hand, the lower part **92** may extend from the center of the upper part **91** in the lower direction. The lower part **92** may be smaller than the inner diameter of the seating portion **65** and extend through the seating portion **65** in the lower direction. The lower part **92** may extend to a length such that the shaft **90** and the upper supporter **80** can be engaged with each other when the filter **40** is coupled to the head **60**.

[0161] A water outlet guide portion **965b** may be formed on one side surface of the lower part **92**. The water outlet guide portion **965b** may extend vertically from the lower side of the shaft outlet port **962**. For example, the water outlet guide portion **965b** may be formed by cutting a portion of the outer surface of the lower part **92** having a cylindrical shape into a planar shape.

[0162] Therefore, in a state where the shaft **90** is mounted on the head **60**, the water outlet guide portion **965b** of the shaft **90** may be spaced apart from the inner surface of the head **60** and thus the shaft water outflow path **965** which is a flow path of water is formed. Since the lower end of the water outlet guide portion **965b** is located on the upper side of the filter outlet port **862**, the water discharged from the filter outlet port **862** is moved along the water outlet guide portion **965b** in the upper direction, and then passes by the shaft outlet port **962** and the water outlet portion **612** in this order and flows to the water outlet pipe **302**.

[0163] FIG. **15** is a cutaway exploded perspective view illustrating an internal structure of the head viewed from one side according to an embodiment of the invention. FIG. **16** is a cutaway exploded perspective view illustrating the internal structure of the head viewed from the other side.

[0164] FIG. **15** illustrates a longitudinal section in a state where the bypass flow path **95** is switched to be connected to the water inlet portion **611** and the water outlet portion **612**. As shown, a bypass flow path **95** passing through the center of the upper part **91** may be formed on the upper part **91**, and the bypass inlet port **951** and the bypass outlet port **952** may be formed on both sides of the circumference of the upper part **91**, respectively.

[0165] The bypass inlet port **951** and the bypass outlet port **952** may be larger than the sizes of the water inlet port **611a** and the water outlet port **612a**, and may be positioned between a pair of shaft O-rings which are vertically disposed. Therefore, the water passing through the bypass flow path **95** in a state where the bypass flow path **95** is aligned with the water inlet portion **611** and the water outlet portion **612** does not leak to the outside and passes across the head **60**.

[0166] In other words, water introducing into the head **60** passes through the head **60** without passing through the filter **40**, and even in a state where the filter **40** is separated, water does not leak from a side on which the filter **40** is mounted and can be continuously supplied to the water supplying flow path **30**.

[0167] On the other hand, a flow path projecting portion **963** which projects for forming the filtering flow path **96** (described in more detail below) is formed on the inner surface of the bypass flow path **95**. The flow path projecting portion **963** may be formed at the center of the inside portion of the shaft **90** and project from the bottom of the bypass flow path **95** but may be formed not to shield the bypass flow path **95**.

[0168] Both ends of the flow path projecting portion **963** may be inclined or rounded so that a decrease in the flow velocity caused by the flow path projecting portion **963** can be minimized when water flows through the bypass flow path **95**.

[0169] FIG. **16** illustrates a longitudinal section in a state where the filtering flow path is shifted to be connected with the water inlet portion and the water outlet portion. As shown, the filtering flow path **96** may be connected to the water inlet portion **611** and the water outlet portion **612** according to the rotation of the shaft **90**.

[0170] At this time, the shaft inlet port **961** and the shaft outlet port **962** are in contact with the water inlet port **611a** and the water outlet port **612a**, respectively, and the outsides of the shaft inlet port **961** and the shaft outlet port **962** and the outsides of the water inlet port **611a** and the water

outlet port **612a** are capable of being fully sealed by the sealing member **94**. An inner pipe **97** may be formed on the inside of the lower part **92**, and the inner pipe **97** can be connected to the supporter extending portion **85**.

[0171] Accordingly, the water which flows into through the shaft **90** can be supplied to the inside portion of the filter **40** through the upper supporter **80** and discharged to the shaft **90** through the upper supporter **80** after the water is purified by the filtering member **44**. In other words, the water which flows into the head **60** can be purified through the filter **40** and then discharged through the head **60**.

[0172] On the other hand, the filtering flow path **96** may include a shaft water inlet flow path **964** and a shaft water outlet flow path **965**.

[0173] The shaft water inlet flow path **964** may include a horizontal portion **964a** that extends from the shaft inlet port **961** to the center of the shaft **90** and a vertical portion **964b** that extends from an end portion of the vertical portion **964a** in the lower direction. The vertical portion **964b** may be formed by the inner pipe **97**.

[0174] The outer surface of the inner pipe **97** may be spaced apart from the inner surface of the lower part **92** to form a spacing space **971**. The distance of the spacing space **971** may correspond to the thickness of the supporter extending portion **85**. Therefore, when the filter **40** is mounted, the upper end of the supporter extending portion **85** can be inserted into an inside of the spacing space **971**.

[0175] The length of the inner pipe **97** in the vertical direction may be shorter than the length of the outer surface of the lower part **92** in the vertical direction. The inner pipe **97** and the supporter extending portion **85** may be connected together on the inner surface of the lower part **92**. To this end, a second connecting portion **972** may be formed on the lower end of the inner pipe **97**.

[0176] The second connecting portion **972** may have a shape corresponding to the first connecting portion **851**. The second connecting portion **972** may be inserted into the first connecting portion **851** so that the shaft **90** and the upper supporter **80** can be rotated together.

[0177] The shaft water inlet flow path **964** may communicate with the filter inlet flow path **871**. The water for purification can be supplied to the filtering member **44** by the coupling together the inner pipe **97** and the supporter extending portion **85**.

[0178] The shaft water outlet flow path **965** may include a water outlet guide portion **965b** formed on an outer surface of the lower part **92**, and a water outlet connecting portion **965a** formed on the upper part **91**.

[0179] The upper end of the water outlet guide portion **965b** may pass through the lower surface of the upper part **91** and then communicate with the water outlet connecting portion **965a**. The water outlet connecting portion **965a** may connect the water outlet guide portion **965b** and the shaft outlet port **962** at the inside of the upper part **91**.

[0180] Accordingly, the purified water which is discharged from the filter outlet port **862** can move along the water guide portion **965b** in the upper direction and be discharged to the shaft outlet port **962** through the water outlet connecting portion **965a**. The purified water which is discharged to the shaft outlet port **962** may be discharged through the water outlet portion **612**.

[0181] Accordingly, for example, in a state where the filtering flow path **96** is connected to the water inlet portion **611** and the water outlet portion **612**, the water which flows into the head **60** through the water inlet pipe **301** is supplied into the inside portion of the filter **40** and then is capable of being purified, and may flow again from the filter **40** to the head **60** and be discharged to the water outlet pipe **302**.

[0182] On the other hand, the body seating portion **65** may be formed on the inner circumferential surface of the upper body **64**. The lower surface of the upper part **91** may be seated on the body seating portion **65** when the shaft **90** is mounted.

[0183] FIG. **17** is a perspective view illustrating the head viewed from below according to an embodiment of the invention. FIG. **18** is a partially cutaway perspective view illustrating the head.

[0184] With reference to FIGS. 15-18, a rotating guide **651** may be formed on the body seating portion **65**. The rotating projection **921** may be positioned on an inside of the rotating guide **651**. The rotating guide **651** may be formed on the body by cutting the body at an angle of 180 degrees with respect to the center of the head body **61**. However, it is not limited thereto.

[0185] Since a pair of rotating projections **921** are disposed at an angle of 90 degrees, in a case where the shaft **90** may be rotated by the angle of 90 degrees, the rotating projections **921** is capable of being stopped by stoppers **652** which are formed on both ends of the rotating guide **651**.

[0186] The water inlet portion **611** and the water outlet portion **612** may be selectively connected to the filtering flow path **96** or the bypass flow path **95** at a position which is stopped by the stopper **652**. Therefore, when the user rotates the filter **40** to a point where the filter **40** is no longer rotated in one direction even without rotating the filter **40**, the flow path is capable of being selected and accurately connected.

[0187] A flow path cutout portion **653** may be formed on one side of the body seating portion **65**. The flow path cutout portion **653** may be formed on the seating portion **65** in a position facing the rotating guide **651** and may be formed on the lower side of the water outlet port **612a** in the vertical direction.

[0188] Therefore, the flow path cutout portion **653** may be disposed at a position which is the same as that of the water outlet guide portion **965b** in a state where the shaft **90** is rotated so that the filtering flow path **96** is connected to the water inlet portion **611** and the water outlet portion **612**. The flow path cutout portion **653** may have the same (or substantially the same) width as the water outlet guide portion **965b**. Therefore, the water outlet guide portion **965b** and the flow path cutout portion **653** can be in contact with each other to form a flow path through which the purified water is capable of flowing in the upper direction.

[0189] On the other hand, a pair of the coupling grooves **631** and a plurality of the supporting ribs **634** may be formed on the inner surface of the lower body **63**. When the filter **40** is mounted on the head **60** and then rotated so that the filtering flow path **96** is connected to the water inlet portion **611** and the water outlet portion **612**, the coupling projection **433** can be inserted into the coupling groove **631**.

[0190] The coupling groove **631** may be exposed to the outside portion through the opening portion **632**, and the coupling state of the restraining projection **433b** and the restraining groove **631a** can be checked through the opening portion **632**. The position of the restraining groove **631a** may be formed at a position in which the restraining projection **433b** and the restraining groove **631a** may be engaged to be restrained with each other in a state where the coupling projection **433** is fully rotated. Therefore, in a state where the filter **40** is inserted into the inside of the head **60** and then fully rotated, the coupling projection **433** is capable of being restrained in the inside of the coupling groove **631** and thus random separation of the filter can be prevented.

[0191] On the other hand, the coupling groove **631** may be formed by a first guide projecting portion **635** which projects from the inner surface of the lower body **63**. The restraining groove **631a** may be formed on the upper surface of the first guide projecting portion **635**. The first guide projecting portion **635** may be formed from the opened end of the lower body **63** to the coupling groove **631** and provide a surface on which the coupling projection **433** can be seated.

[0192] Therefore, when the filter **40** is inserted into the head **60**, the coupling projection **433** cannot be inserted into some sections of the opening of the lower surface of the head **60** due to the interference of the first guide projecting portion **635**. The coupling projection **433** can be inserted through the section which is not interfered with the first guide projecting portion **635**, so that the filter **40** can be prevented from being erroneously mounted.

[0193] A first groove guide portion **633a** for guiding the coupling projection **433** to the entrance of the adjacent coupling groove **631** may be formed on one side surface of the first guide projecting portion **635**. The first groove guide portion **633a** may have a predetermined inclination and be in contact with the projection guide portion **433a** of the coupling projection **433** and thus guide the

rotational movement of the coupling projection **433** in one direction when the coupling projection **433** is inserted.

[0194] A second guide projecting portion **636** may be formed on one side which is spaced apart from the end portion of the first groove guide portion **633a** along the inner surface of the lower body **63** by a predetermined distance. A second groove guide portion **633b** may be inclined on the second guide projecting portion **636** so that the coupling projection **433** which passes by the first groove guide portion **633a** moves along the second groove guide portion **633b** so as to be guided and moved to the entrance of the coupling groove **631**. The second groove guide portion **633b** may extend from one side away from the first groove guide portion **633a** to the entrance of the coupling groove **631**.

[0195] Therefore, when the filter **40** is inserted into the opened lower surface of the lower body **63** after the coupling projection **433** is positioned at a position corresponding to the inserting display portion **613** when the filter **40** is mounted, the projection guide portion **433a** slides along the first groove guide portion **633a** and then slides along the second groove guide portion **633b** and is inserted into the inside of the coupling groove **631**.

[0196] In a case where the user inserts the coupling projection **433** of the filter **40** by aligning the coupling projection **433** in a process of the filter **40** being inserted into the inside of the head **60**, the filter **40** can be coupled while being smoothly rotated by the first groove guide portion **633a** and the second groove guide portion **633b**.

[0197] The shaft **90** and the upper supporter **80** are integrally coupled together in a state where the filter **40** is fully inserted and then the coupling projection **433** is positioned at the entrance of the coupling groove **631**. When the filter **40** is further rotated so that the coupling projection **433** is fully inserted into the inside of the coupling groove **631**, the shaft **90** is rotated together with the filter **40** and the filtering flow path **96** is rotated to be connected to the water inlet portion **611** and the water outlet portion **612**.

[0198] To this end, a second connecting portion **972** may be formed at the lower end of the shaft **90**, that is, at the lower end of the inner pipe **97**. A pair of second connecting portions **972** may have the same shape at a position which faces each other, and both sides of the lower end of the inner pipe **97** may be formed by a cutting operation. However, it is not limited to thereto.

[0199] Specifically, the lower end of the inner pipe **97** may include the pair of second connecting portions **972** and a pair of pipe cutout portions **973** formed between the second connecting portions **972**. The pair of second connecting portions **972** may have a width which is gradually narrowed in the lower direction.

[0200] A second inclined surface **974** may be formed at both side ends of the pair of second connecting portions **972** and the second inclined surface **974** may be formed to have an inclination corresponding to the first inclined surface **853**.

[0201] The second inclined surface **974** may be inserted along the first inclined surface **853** in a process of the filter **40** being rotatably inserted into the head **60**. When the filter **40** is fully inserted into the head **60**, the second connecting portion **972** may be matched with the first connecting portion **851** and the projecting portion **852** may be matched with the pipe cutoff portion **973**. Therefore, the first inclined surface **853** and the second inclined surface **974** can be in close contact with each other.

[0202] In addition, the second inclined surface **974** may be inclined in the direction of rotation of the filter **40** as the first inclined surface **853**. Therefore, when the filter **40** is further rotated in a state where the rotation of the shaft **90** is restricted by the stopper **652**, the force in the rotating direction is acted to the second inclined surface **974** and the first inclined surface **853** and the first inclined surface **853** moves along the second inclined surface **974** so that the filter **40** can be more easily separated.

[0203] On the other hand, a plurality of the supporting ribs **634** may be formed on the inner surface of the lower body **63**, and the outer surface of the filter inserting portion **431** can be supported by

the supporting ribs **634**.

[0204] Hereinafter, the operation of the water purifying apparatus according to an embodiment of the invention having the structure described above is described.

[0205] FIG. **19** is a cutaway perspective view illustrating a state where of the filter and the head are separated from each other according to an embodiment of the invention. FIG. **20** is an enlarged view of portion A of FIG. **19**, and FIG. **21** is a view illustrating a shaft position in a state where the filter and the head are separated from each other.

[0206] As illustrated in the drawings, the bypass flow path **95** may be connected to the water inlet portion **611** and the water outlet portion **612** in a state where the filter **40** is not coupled to the head **60**.

[0207] Therefore, the water which flows into through the water inlet portion **611** flows into the bypass inlet port **951** through the water inlet port **611a** and flows along the bypass flow path **95**. The water is discharged to the water outlet portion **612** through the bypass outlet port **952** and the water outlet port **612a**. In other words, the water which flows into the water inlet portion **611** passes through the head **60** without being subjected to a purification process, and is supplied directly to the ice maker **24** or the dispenser **23** through the water supplying flow path **30**.

[0208] Such a state may correspond to a state where the filter **40** is separated for replacement of the filter **40**, or may correspond to a situation where the service which is related to the cleaning of a pipeline or other maintenance is performed. In addition, even in a case where at least a portion of the plurality of filters **40** are not used, or where purification of water is not required, it is capable of corresponding to such a state. Moreover, even in a case where the filter **40** is fully separated, no problem occurs in use of the refrigerator **1**.

[0209] On the other hand, with reference to FIG. **21**, when the disposition of the shaft **90** is described in a state where the filter **40** is not mounted, the first rotating projection **921a** of the pair of rotating projections **921** is in a state of being in contact with the stopper **652** of one side of the rotating guide **651**. In this state, the water outlet guide portion **965b** and the flow path cutout portion **653** are maintained a state of being shifted by an angle of about 90 degrees with each other.

[0210] FIG. **22** is a view illustrating a state where the filter is inserted into an inside of the head according to an embodiment of the invention. FIG. **23** is a view illustrating a state where the filter is fully inserted into the inside of the head.

[0211] As illustrated in the drawings, the filter inserting portion **431** is inserted into the opening of the lower surface of the head **60** to mount the filter **40**. At this time, the position of the coupling projection **433** is capable of being aligned and inserted by referring to the inserting display portion **613** formed on the outside of the head **60**.

[0212] The coupling projection **433** slides along the first groove guide **633a** and the second groove guide **633b** in a process of the insertion of the filter **40**. Accordingly, the filter **40** can be smoothly rotated along with insertion.

[0213] When the filter **40** is inserted by a predetermined depth, the insertion of the second connecting portion **972** into the inside of the first connecting portion **851** proceeds, as illustrated in FIG. **21**. At this time, the second connecting portion **972** and the first connecting portion **851** are capable of being slid while the second inclined surface **974** and the first inclined surface **853** are in contact with each other, respectively. Accordingly, the second connecting portion **972** may be guided to the inside of the first connecting portion **851**.

[0214] In other words, when the filter **40** is inserted and mounted, the filter **40** is smoothly rotated in the process of inserting the filter **40** into the head **60** by the first groove guide **633a** and the second groove guide **633b**. The shaft **90** and the supporter extending portion **85** can be fully matched by sliding the second inclined surface **974** and the first inclined surface **853** in the process of rotation, as illustrated in FIG. **23**.

[0215] As illustrated in FIG. **23**, the coupling projections **433** may be located on the inner side or the entrance side of the coupling grooves **631** in a case where the shaft **90** and the supporter

extending portion **85** are fully matched with each other. In other words, the filter **40** corresponds to a state where the filter **40** is simply inserted before the filter **40** is fully coupled with the head. In this state, the shaft **90** is not rotated, and the bypass flow path **95** is in a state of being connected to the water inlet portion **611** and the water outlet portion **612**.

[0216] Moreover, in the state illustrated in FIG. **23**, the user is capable of further rotating the filter **40** in the rotation advancing direction and is rotated by about 90 degrees so that the coupling projection **433** is fully inserted into the inside of the coupling groove **631**.

[0217] FIG. **24** is a view illustrating a state where the filter is rotated for coupling in a state where the filter is fully inserted into the inside of the head according to an embodiment of the invention.

FIG. **25** is an enlarged view of portion B in FIG. **24**, and FIG. **26** is a view illustrating a shaft position in a state where the filter is coupled to the head.

[0218] In a state of being illustrated in FIG. **23**, when the filter **40** is further rotated by an angle of 90 degrees so that the coupling projection **433** and the coupling groove **631** are rotated to be fully coupled with each other, the shaft **90** is also rotated along with the rotation of the filter and thus is in a state of being illustrated in FIG. **26**.

[0219] Specifically, when the filter **40** is rotated in a state where the second connecting portion **972** is inserted into the first connecting portion **851** and thus fully coupled together, the shaft **90** rotates along with the filter **40**.

[0220] The shaft **90** may be further rotated by an angle of about 90 degrees until the second rotating projection **921b** reaches the position of the stopper **652**. When the shaft **90** is fully rotated, the filtering flow path **96** is connected to the water inlet portion **611** and the water outlet portion **612**. Of course, the inner pipe **97** and the supporter extending portion **85** maintains a connected state with each other and thus the original water and the purified water is capable of flowing in and out between the head **60** and the filter **40**. In addition, the coupling projection **433** is in a state of being fully inserted into and thus is coupled with the coupling groove **631** and the filter **40** is operated to be rotated in a direction which is opposite to the coupling direction by the user and thus the filter **40** is maintained in a state of coupling with the head **40** until the filter **40** is separated from the head **60**.

[0221] Moreover, in the state illustrated in FIG. **24**, the water flowing in through the water inlet portion **611** flows along the shaft water inlet flow path **964** through the water inlet port **611a** and the shaft inlet port **961**. In other words, water flowing along the horizontal portion **964a** and the vertical portion **964b** passes by the inner pipe **97**, flows into the inside of the supporter extending portion **85** and flows along the first filter inlet flow path **872** in the lower direction. Then, water is branched by the second filter inlet flow path **873** and flows into the space between the inner surface of the housing **41** and the filtering member **44**.

[0222] The water flowing into the hollow **441** of the inside of the filtering member **44** from the outside of the filtering member **44** may be purified in the process of passing through the filtering member **44**. The purified water of the inside of the filtering member **44** flows along the inside of the first extending portion **86** in the upper direction and is discharged from the filter outlet port **862** which is disposed on both sides of the upper end of the first extending portion **86** in the upper direction.

[0223] At this time, a space in which water purified by the outer surface of the first extending portion **86** and the lower part **92** inserted into the inside of the filter inserting portion **431** flows to the head **60** side may be formed in the inside of the filter inserting portion **431**. On the other hand, as illustrated in FIG. **24**, in a state where the filtering flow path **96** is connected to the water inlet portion **611** and the water outlet portion **612**, the water outlet guide portion **965b** and the flow path cutout portion **653** are positioned at the same position with each other and thus the purified water is capable of flowing to the upper part **91** side.

[0224] Accordingly, the purified water flows along the water outlet guide portion **965b** formed in the lower part **92** in the upper direction and flows into the inside of the upper part **91** and thus

passes through the shaft outlet port **962** and the water outlet portion **612** in this order, and then is discharged to the water outlet portion **612**. The water outlet pipe **302** of the water outlet portion **612** forms a portion of the water supplying flow path **30** to supply the purified water to the dispenser **23** and the ice maker **24**.

[0225] Moreover, in the state illustrated in FIG. **23**, in a state where the filter **40** should be removed because of reach of period of replacement of the filter **40** or other maintenance, the filter **40** is first rotated in a direction which is opposite to the coupling direction.

[0226] The coupling projection **433** may be moved in a direction away from the coupling groove **631** according to the rotation of the filter **40** and the supporter extending portion **85** rotates the shaft **90**. As illustrated in FIG. **20**, when the filtering flow path **96** is closed by the shaft **90** being rotated by 90 degrees and then the bypass flow path **95** is connected, the rotating projection **921** contacts the stopper **652** and thus is restricted the rotation of the shaft **90**.

[0227] In this state, when a force is applied to further rotate the filter **40**, the shaft **90** cannot be further rotated based on interference with the stopper **652**. Accordingly, a torsion moment is applied to the first inclined surface **853** and the second inclined surface **974** so that the second inclined surface **974** is smoothly separated along the first inclined surface **853** while being slid. Further, the coupling projection **433** which is escaped from the coupling groove **631** passes by the second groove guide portion **633b** and the first groove guide portion **633a** in this order to allow the filter **40** to be separated from the head **60**.

[0228] In the disclosed embodiments of the invention, so as to facilitate the understanding of the invention, as an example, the water purifying apparatus **17** is described as being mounted on the refrigerator **1**. However, it is understood that the water purifying apparatus **17** described herein can likewise be applied to the water purifier which is generally used and the entire device which is capable of purifying water by a filter exchanging manner.

[0229] The water purifying apparatus according to the disclosed embodiments and the refrigerator including the water purifying apparatus have at least the following effects.

[0230] According to the disclosed embodiments of the invention, in a case where the filter is separated from the head for replacement or maintenance of the filter, the bypass flow path connects to the water inlet portion and the water outlet portion so that the flow path is capable of being connected without generating leakage. Since the water supplying flow path is capable of being maintained in a connected state by the bypass flow path, continuous supply of water to the dispenser or the ice maker becomes possible, and service work such as the cleaning of the flow path becomes possible. Therefore, convenience of use of the water purifying apparatus is improved.

[0231] In addition, according to the disclosed embodiments of the invention, the filtration flow path is connected to the water inlet portion and the water outlet portion during the rotation operation for mounting the filter, so that purified water is capable of being supplied. When rotation for the separation of the filter is operated, the bypass flow path is connected to the water inlet portion and the water outlet portion so that water is capable of being continuously supplied even at the removal of the filter. In other words, the flow path can be switched without additional operation by the rotation operation of mounting and separating the filter. Thus, convenience of use of the water purifying apparatus is further improved.

[0232] In addition, according to the disclosed embodiments of the invention, the exact position is capable of being displayed on the head into which the filter is inserted so that the filter is capable of being mounted accurately. In addition, since a groove guide which is in contact with the coupling projection is formed at the inside of the head so as to be inclined, the filter is capable of being smoothly rotated and thus inserted into the coupling groove only by inserting the filter at an exact position. In addition, since the flow paths of the head and the filter are capable of being connected at the same time by the coupling of the coupling projections with the coupling grooves. Thus, convenience of use of the water purifying apparatus is further improved.

[0233] In addition, according to the disclosed embodiments of the invention, the first connecting

portion and the second connecting portion which are formed at the ends of the shaft and the upper supporter have a structure which are matched with each other and have an inclination in the axial direction and thus when the filter is inserted into the head, since the inclinations are in contact with each other, the filter is guided to be inserted in the exact position. In addition, when the filter is rotated to separate the filter, the inclined surfaces of the first connecting portion and the second connecting portion are capable of being easily separated from each other in a section in which the rotation of the shaft is restricted. Thus, convenience of use of the water purifying apparatus is further improved.

[0234] In addition, according to the disclosed embodiments of the invention, since the flow path is capable of switching by only the rotation of the shaft without a separate valve structure and an elastic member for switching the flow path, the configuration of the product is capable of being simplified. Thus, productivity can be improved and manufacturing cost can be reduced.

[0235] In addition, according to the disclosed embodiments of the invention, malfunction of the water purifying apparatus is prevented or substantially minimized due to a less complex operation structure, reliability of the water purifying apparatus is assured, and the durability of the water purifying apparatus is improved.

[0236] In addition, according to the disclosed embodiments of the invention, since a portion of the filtering flow path projects to the inside of the bypass flow path, so that the structure of the shaft is capable of being made compact. Thus, the capacity loss of the refrigerator when the water purifying apparatus is installed in the refrigerator can be minimized by the size of the head on which the shaft is mounted, and the water purifying apparatus can have a more compact structure.

[0237] The foregoing description is merely illustrative of the technical idea of the present invention and various changes and modifications may be made without departing from the essential characteristics of the present invention by those having ordinary skill in the art to which the present invention belongs. Therefore, the embodiments disclosed in the present invention are intended to illustrate rather than limit the scope of the present invention, and the scope of the technical idea of the present invention is not limited by these embodiments.

Claims

1. A water filter comprising: a filtering member; a filter housing that accommodates the filtering member therein; and an upper supporter accommodated within an accommodating space defined at an upper portion of the filter housing, wherein the upper supporter is configured, based on the filter housing being rotated, to rotate together with the filter housing, wherein the upper supporter comprises: a supporter accommodating portion that is coupled to an upper end of the filtering member to support the filtering member in the filter housing, a supporter extending portion comprising a filter inlet flow path configured to receive water through an upper opening of the filter housing, wherein the supporter extending portion comprises: a connecting portion provided at an upper end of the supporter extending portion, the connecting portion comprising an interior recess extending from an upper portion of the filter inlet flow path and configured for coupling to a shaft; a projecting portion formed at an entrance of the upper portion of the filter inlet flow path and configured for rotating the shaft; and a sealing member disposed on an outer surface of the supporter extending portion to provide a seal between the outer surface of the supporter extending portion and an inside of the shaft, the sealing member spaced from an inner surface of the filter housing.
2. The water filter of claim 1, wherein the filter housing comprises a housing body and a housing cap at an upper portion of the housing body, and wherein the accommodating space is defined in the housing cap.
3. The water filter of claim 2, wherein the housing cap comprises a housing fastening portion provided at an inner surface of the housing cap, and wherein the supporter extending portion

further comprises a supporter fastening portion configured to couple to the housing fastening portion of the housing cap.

4. The water filter of claim 3, wherein the supporter fastening portion is disposed below the sealing member.

5. The water filter of claim 3, wherein the supporter fastening portion is disposed below the sealing member and above the supporter accommodating portion.

6. The water filter of claim 3, wherein the housing fastening portion includes at least two housing fastening portions that face each other.

7. The water filter of claim 6, wherein the supporter fastening portion includes at least two supporter fastening portions that face each other.

8. The water filter of claim 7, wherein the supporter fastening portion is configured to become engaged with the housing fastening portion to thereby couple the upper supporter to the housing cap.

9. The water filter of claim 8, wherein a width of the projection portion along the circumferential direction increases from an upper end of the projection portion toward a lower end.

10. The water filter of claim 1, wherein the filter housing further includes a filter insertion portion provided at an upper portion of the filter housing, and wherein the supporter extending portion is located within the filter insertion portion.

11. The water filter of claim 10, wherein the supporter extending portion includes: a first extending portion that extends vertically upward from the supporter accommodating portion, and a second extending portion that extends vertically upward from the first extending portion.

12. The water filter of claim 11, wherein a first outer diameter of the first extending portion is greater than a second outer diameter of the second extending portion.

13. The water filter of claim 12, wherein the filter inlet flow path extends vertically downward through the second extending portion and the first extending portion, the filter inlet flow path is configured to guide water from a water supply source toward the filter member.

14. The water filter of claim 13, wherein the connecting portion is provided at the upper end of the second extending portion.

15. The water filter of claim 10, wherein, based on the filter housing and the upper supporter being engaged with each other, an uppermost side of the connecting portion is positioned at a vertical position lower than an uppermost side of the upper end of the filter insertion portion.

16. The water filter of claim 11, wherein the first extending portion extends vertically upward from an upper surface of a supporter stepped portion that is stepped relative to the supporter accommodating portion.

17. The water filter of claim 6, wherein each of the at least two housing fastening portions includes an inclined upper surface.

18. The water filter of claim 7, wherein each of the at least two supporter fastening portions including a hook portion that is configured to engage with a corresponding one of the at least two housing fastening portions to thereby couple the upper supporter to the housing cap.

19. The water filter of claim 16, wherein the supporter stepped portion defines a second filter inlet flow path therein.

20. A water filter comprising: a filtering member; a filter housing that accommodates the filtering member therein, wherein the filter housing comprises a housing body and a housing cap at an upper portion of the housing body; and an upper supporter accommodated within an accommodating space defined in the housing cap, wherein the upper supporter and the housing cap are configured, based on the housing body being rotated, to rotate together with the housing body, wherein the housing cap comprises a housing fastening portion provided at an inner surface of the housing cap, and wherein the upper supporter comprises: a supporter accommodating portion that is coupled to an upper end of the filtering member to support the filtering member in the housing body, a supporter extending portion comprising a filter inlet flow path configured to receive water through

an upper opening of the housing cap, wherein the supporter extending portion comprises: a supporter fastening portion configured to couple to the housing fastening portion of the housing cap; a connecting portion provided at an upper end of the supporter extending portion, the connecting portion comprising an interior recess extending from an upper portion of the filter inlet flow path and configured for coupling to a shaft; a projecting portion formed at an entrance of the upper portion of the filter inlet flow path and configured for rotating the shaft; and a sealing member disposed on an outer surface of the supporter extending portion to provide a seal between the outer surface of the supporter extending portion and an inside of the shaft, the sealing member spaced from an inner surface of the housing cap; wherein the supporter fastening portion is disposed below the scaling member and above the supporter accommodating portion.
